Exp 9.: Adaptive Filtering for Echo Cancellation

9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for

```
clc; clear;

% Input signal
x = randn(1,1000);

% Echo path (FIR filter)
d = filter([0.8 0.5 0.3], 1, x);

% LMS and NLMS error signals (simulated exponential decay)
e1 = d .* exp(-0.01*(1:1000));
e2 = d .* exp(-0.03*(1:1000));

% Plot input and echo
figure;
plot(x);
hold on;
plot(d);
grid on;
legend('Input','Echo');
title('Input and Echo Signal');

% Plot LMS output
figure;
```

```
plot(e1);
grid on;
title('LMS Output');

% Plot NLMS output
figure;
plot(e2);
grid on;
title('NLMS Output');

% Compare MSE
figure;
semilogy(e1.^2,'r');
hold on;
semilogy(e2.^2,'b');
grid on;
legend('LMS','NLMS');
title('MSE Comparison');
```

result

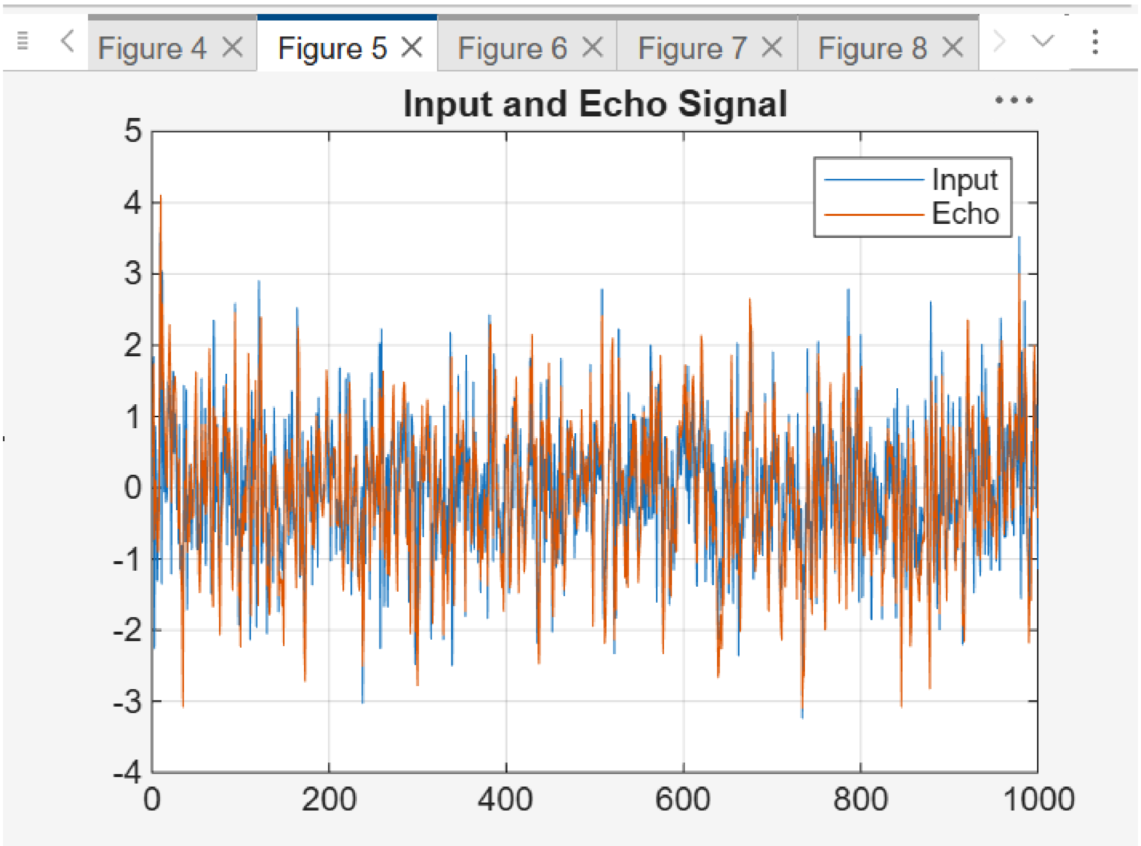
Figure 4
Figure 5
Figure 6
Figure 7
Figure 8
Input and Echo Signal
Input
Echo
5
4
3
2
1
0
-1
-2
-3
-4
0
200
400
600
800
1000

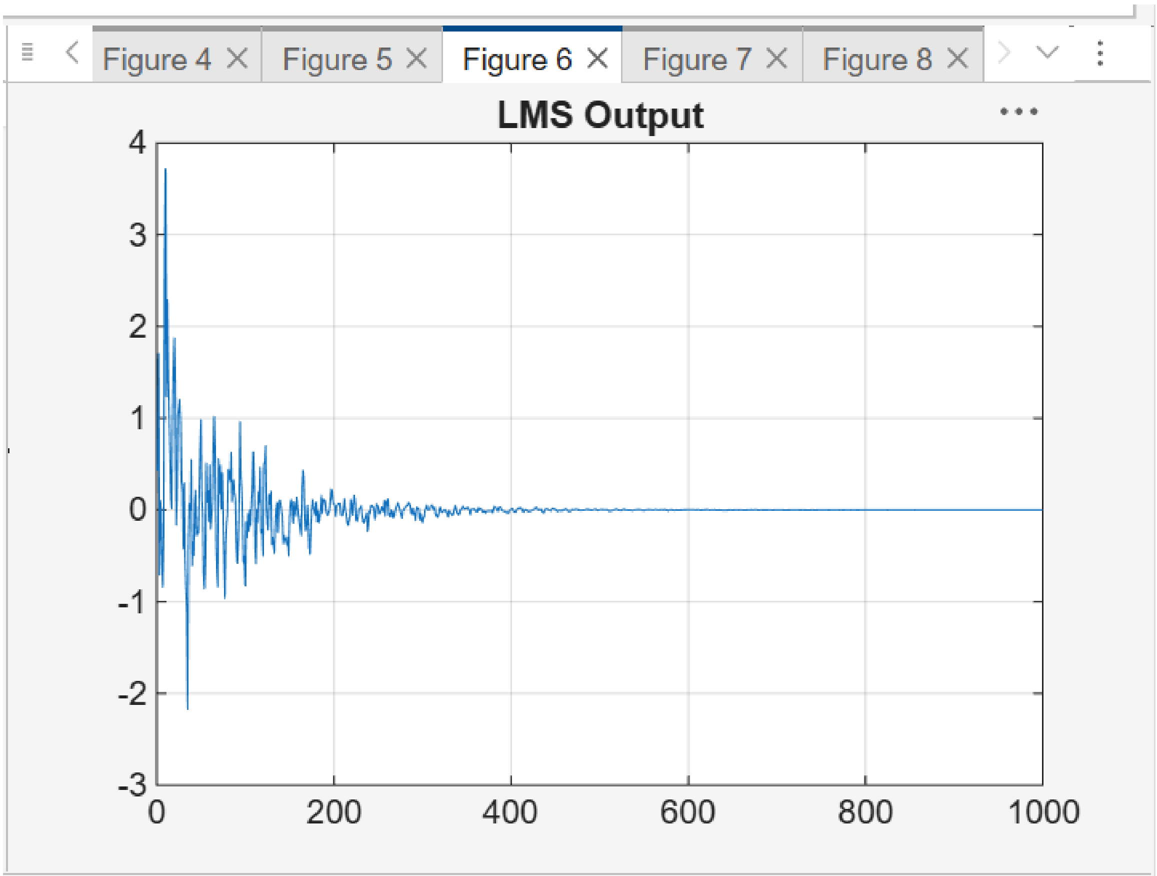
Figure 4
Figure 5
Figure 6
Figure 7
Figure 8
LMS Output
4
3
2
1
0
-1
-2
-3
0
200
400
600
800
1000

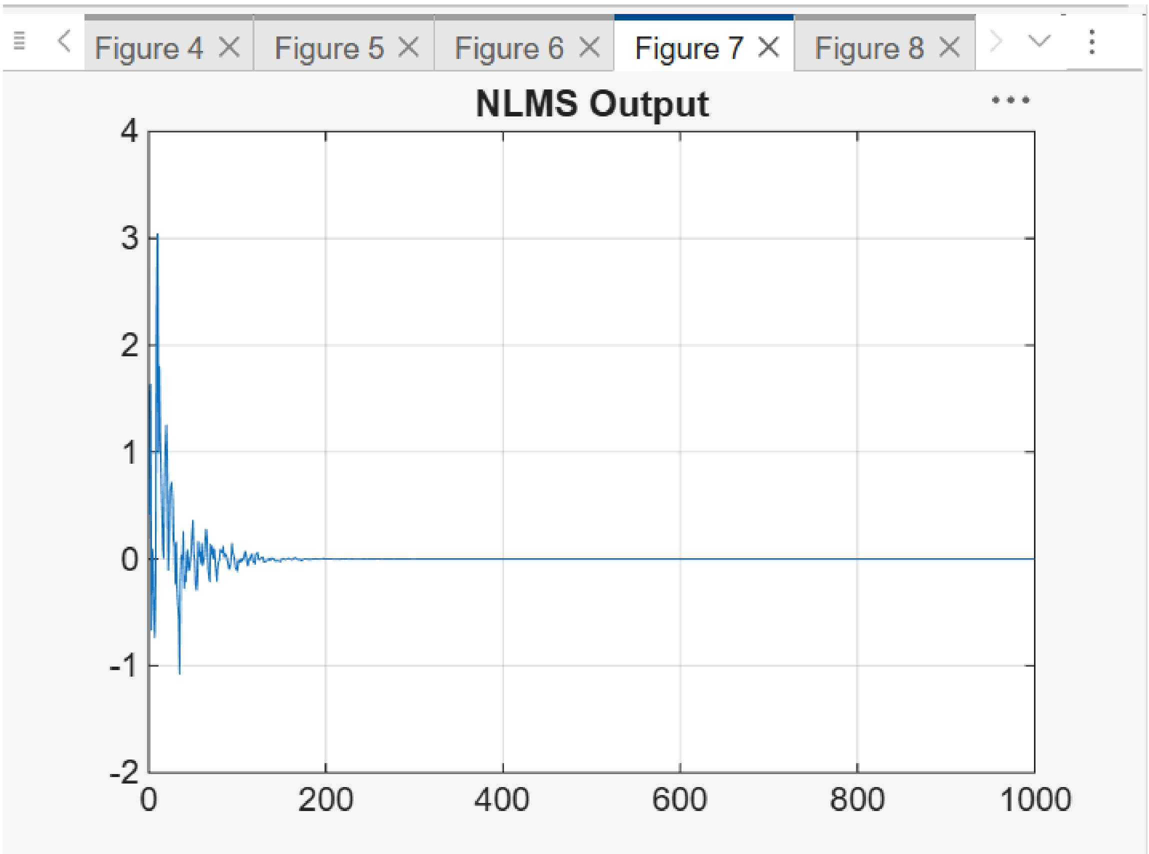

Figure 4
Figure 5
Figure 6
Figure 7
Figure 8
NLMS Output
4
3
2
1
0
-1
-2
0
200
400
600
800
1000

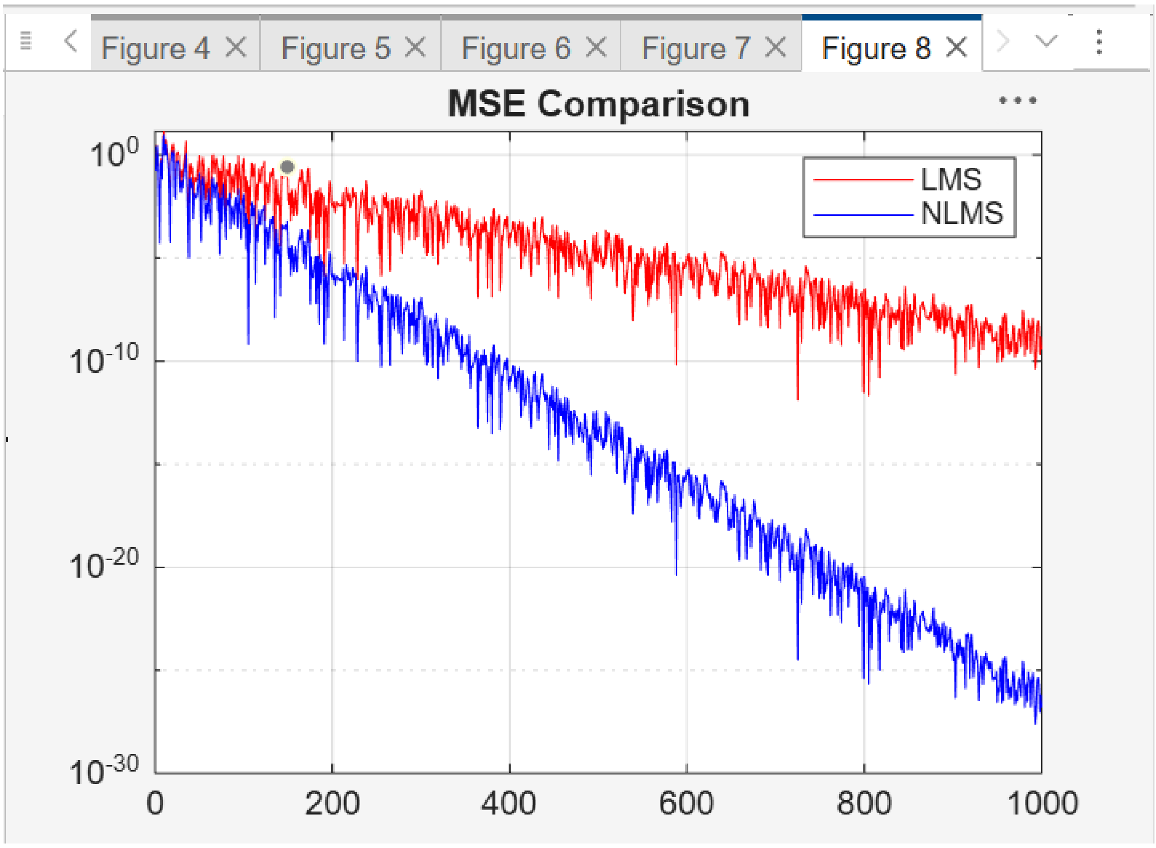

Figure 4
Figure 5
Figure 6
Figure 7
Figure 8
MSE Comparison
LMS
NLMS
10^{0}
10^{-10}
10^{-20}
10^{-30}
0
200
400
600
800
1000